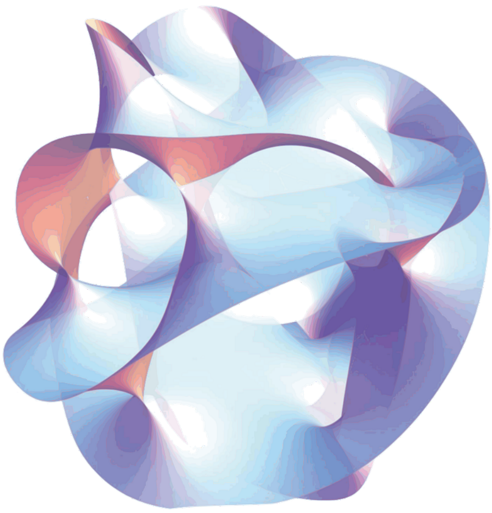
Spatial dimension compactification in general relativity and duality symmetry

Adam Franchot–Fowler

Internship carried out from 08/06/2026 to 24/07/2026



Internship tutors : M. Camille Eloy

Referent teacher : M. Stéphane Douin

Éducational institution : Université Paris-Saclay / Magistère de phyisque fondamentale d'Orsay

Internship host company : Laboratoire de Physique (UMR CNRS 5672)ENS de Lyon 46, allée d'Italie F-69364 LYON CEDEX 07, FRAN

Acknowledgment

Notations and abreviations

For the abreviations we wil use

- GR for general relativity

We will during this intership use the vielbein formalism. We will use the same notation as in [1] which is

- Greek hatted letter for curved indices $(\hat{\mu}, \hat{\nu}, \hat{\rho}, \dots)$
- Latin hatted letter for flat indices $(\hat{p}, \hat{m}, \hat{n}, \dots)$
- For space-time indices taking D values, late greek letters are used in the curved case $(\mu, \nu, \rho, \dots)$ and Latin letters in the flat case (p, m, n)
- For internal indices taking E values, early Greek letters in the curved case $(\alpha, \beta, \gamma, \dots)$ and Latin letters in the flat case (a,b,c)

THe general coordinate $x^{\hat{\mu}}$ will thereofre be made of space time coordinate x^{μ} and internal coordinate y^{α}. The flat lorenz

Contents

1 Introduction

Since the 19th-century unification of electricity and magnetism, and the later development of classical and quantum field theory, physicists have been looking for a single mathematical framework that could describe all fundamental interactions at once. A lot of progress has been made for the electromagnetic, weak and strong forces, which are now successfully unified in the Standard Model. Gravity, however, remains the main stumbling block. Unlike the other forces, it resists quantization in a straightforward way, and its description by General Relativity [2] sits completely apart from the quantum field-theoretic language we use for everything else.

One of the earliest attempts to change this was proposed by Kaluza in 1921 [3] and later refined by Klein:
by adding one extra spatial dimension compactified on a small circle, five-dimensional General Relativity naturally reduces to four-dimensional Einstein gravity coupled to electromagnetism. Even if the original theory had problems, this Kaluza-Klein mechanism opened a whole new way of thinking about unification, and it is now at the heart of modern approaches like supergravity [4] and string theory.

2 General relativity theoretical reminder

Here you will find some GR reminder, which hopefully will make it easier to understand the mathematical foundation on which this internship was build

2.1 mathematical basis

Definition 1 *Let V be an R-vector space of dimension N, we define the tensor of order* $(r, s) \in [\![0, 1]\!]^2$ *on* $V^{(r,s)} \overset{def}{=} V^{\otimes r} \otimes V^{\otimes s}$ *as a being a multilinear application on* $V^{\star r} \times V^s$ *to* $\mathbb{R}$

Definition 2 *We will call n-dimension* $(n \in \mathbb{N}^*)$ *manifold every Hausdorff topological space* $(\mathcal{M}, \tau)$ *localy homeomorphic to* $\mathbb{R}^n$*)*

Definition 3 *Let* $(\mathcal{M}, g)$ *be a smooth manifold. A metric tensor is a symmetric, non-degenerate bilinear form* $g_{\mu\nu}$ *on the tangent space* $T_p\mathcal{M}$ *at each point* $p \in \mathcal{M}$*, such that*

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

In General Relativity we use the signature $(-, +, +, +)$.

Definition 4 *In general in RG we use the christofells connexion as it is the sole one wich is mtreci and torsion free. We define it as being*

$$nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\sigma} V^\sigma \tag{1}$$

where the christofells symbols are

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \tag{2}$$

2.2 spin connexion, vielbein and flatspace

3 Lagrangian field theory

3.1 Lagrangian density

In field theory, just as Newton's second law is "volumized" into the Navier-Stokes equations by distributing quantities over space, we replace discrete degrees of freedom with continuous field densities defined at every point in spacetime. Therefore we will rewrite the Lagrangian as the integral over space of a new quantities called Lagrangian density, which will often refered to as just the Lagrangian. Thus we obtain this equation:

$$L = \iiint \mathcal{L}(\eta_\rho, \frac{\partial \eta_\rho}{\partial x^\nu}) dx dy dz \tag{3}$$

Now using esintein convention we can rewritte the action in a new light :

$$S = \int \mathcal{L} dx^\mu \tag{4}$$

3.2 General exemple for scalar and vectorial fields, and result for two forms

In field theory, the Lagrangian has a classical prescription for scalar and vector fields, but no general rule exists for other cases each field type requires its own treatment. For this work, only the scalar, vector, and two-form Lagrangians are needed, along with the metric, which is handled by Einstein's equations.
So for a scalar field we have :

$$\boxed{\mathcal{L} = -\frac{1}{2}\partial_\mu \Phi \partial^\mu \Phi} \tag{5}$$

and for a vectorial field we have

$$\boxed{\mathcal{L} = -\frac{1}{-}F_{\mu\nu}F^{\mu\nu}} \tag{6}$$

where for the field A

$$F_{\mu\nu} = \partial_\mu A^\nu - \partial_\nu A^\mu$$

In a vaccum the einstein equation give us

$$\boxed{\mathcal{L} = R}$$

where R is ricci scalar finally for the lagrangian of a two form fields whe will look in [5] where it is given by the equation :

$$\boxed{\mathcal{L} = g^{\hat{\mu}\mu'}g^{\nu\nu'}g^{\rho\rho'}H_{\mu\nu\rho}H_{\mu'\nu'\rho'}} \tag{7}$$

where for the two form $B_{\nu\rho}$ we have

$$H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + cyc.perm$$

4 Reduction of hilbert-einstein Action in D+E dimension

Here I will do a kalusa klein reduction. Basically we will suppose an universe in D+E dimension where D is 4 and we will try to see how on observater in D will perceive the new dimension.

4.1 Hypothesis

We will consider that the fields don't depend on the internal variable of the world so for any field $\partial_\alpha A^{\hat{\mu}} = 0$. Another thing to note is that originally Kalusa did a reduction on S^1 , but here we will no do that and we will do the reduction on a nondescript space, with E dimensions.

4.2 Rewriting the action

We began our journey with the classical einstein action; with the sole difference being that we are in D+E dimension so we we have as shown in [1]

$$\boxed{S = \frac{-1}{\kappa^2}\int dx^D \int \frac{dy^E}{\rho(E)}\hat{V}\hat{R}} \tag{8}$$

where $\rho(E)$ is the parameter set to normalise 8 and is in practice the invariant volume of the internal space. and we have set κ so that $\frac{\kappa^2}{4\pi}$ is the usual newton constant.
Now working with spin connexion formalism we can rewrite everything in a more understandable manner and we got the action. and get

$$S = \frac{-1}{\kappa^2}\int dx^D \int \frac{dy^E}{\rho(E)}\hat{V}(\hat{\omega}^{\hat{r}}{}_{\hat{s}\hat{t}}\hat{\omega}^{\hat{s}\hat{t}}{}_{\hat{r}} + \hat{\omega}^{\hat{r}}{}_{\hat{r}\hat{t}}\hat{\omega}^{\hat{s}}{}_{\hat{s}}{}^{\hat{t}}) \tag{9}$$

Where the spin connexion are given by

$$\hat{\omega}_{\hat{t}\hat{r}\hat{s}} = -\hat{\Omega}_{\hat{t}\hat{r}\hat{s}} + \hat{\Omega}_{\hat{r}\hat{s}\hat{t}} - \hat{\Omega}_{\hat{s}\hat{t}\hat{r}} \tag{10}$$

and the $\hat{\Omega}$ are given by the inverse vielbein

$$\hat{\Omega}_{\hat{r}\hat{s}\hat{t}} = \frac{1}{2}(\hat{V}^{\hat{\mu}}_{\hat{r}}\hat{V}^{\hat{\nu}}_{\hat{s}} - \hat{V}^{\hat{\mu}}_{\hat{s}}\hat{V}^{\hat{\nu}}_{\hat{r}})\partial_{\hat{\nu}}\hat{V}_{\hat{\mu}\hat{t}} \tag{11}$$

4.3 vielbein anstazs and latin connexion

Now le'ts begin the true work. Firstly as the information of the curved space metric is fully contained withint the vielbein we will do the anstaz of the solution on them.Furthermore according to [1] we can choose a triangular parametrization thanks to the local lorentz invariance in D+E dimension. So our anstazs will be

$$\boxed{\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} \delta^{\gamma} V_{\nu}^{r} & 2\kappa A_{\nu}^{\alpha} \Phi_{\alpha}^{a} \\ 0 & \Phi_{\alpha}^{a} \end{pmatrix}} \tag{12}$$

In this formula $\delta = det\Phi$ and γ is a free paramater, corresponding to an arbitrary Weyl transformation of the D dimensional vielbein.
The vielbein in eq 12 has D dimensional vielbein (also called gravition), E vector fields A_{μ}^{α} and E^2, however the local lorentz invariance in the internal direction, only $\frac{1}{2}E(E+1)$ of the scalars fields correspond to propagating particules.
Using the fact that the theory is invariant under the classical transformation $x'^{\mu} = x^{\mu} + \xi^{m} u(x)$ [1] which can be rewrtintten as being

$$\delta \hat{V}_{\hat{\mu}}^{\hat{r}} = \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}_{\hat{\mu}}^{\hat{r}} + \partial_{\hat{\mu}} \xi^{\hat{\rho}} \hat{V}_{\hat{\rho}}^{\hat{r}} \tag{13}$$

Now we will apply this to eacg block of the vielbein to see if they respect some type of classical field so for the block $\hat{V}_{\mu}^{r}$

$$\begin{aligned}
\delta \hat{V}_{\mu}^{r} &= \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}_{\mu}^{r} + \partial_{\mu} \xi^{\hat{\rho}} \hat{V}_{\hat{\rho}}^{r} \\
&= \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}_{\mu}^{r} + \partial_{\mu} \xi^{\nu} \hat{V}_{\nu}^{r} + \underbrace{\partial_{\mu} \xi^{\alpha} \hat{V}_{\alpha}^{r}}_{\text{Ce terme est nul}} \\
&= \xi^{\hat{\rho}} \partial_{\hat{\rho}} (\delta^{\gamma} V_{\mu}^{r} + \delta^{\gamma} \partial_{\mu} \xi^{\nu} V_{\nu}^{r} \\
&= \delta^{\gamma} (\xi^{\hat{\rho}} \partial_{\hat{\rho}} V_{\mu}^{r} + \partial_{\mu} \xi^{\nu} V_{\nu}^{r}) + \xi^{\hat{\rho}} \partial_{\hat{\rho}} \delta^{\gamma} V_{\mu}^{r}
\end{aligned}$$

Now we can redo this by this time using anither method

$$\begin{aligned}
\delta \hat{V}_{\mu}^{r} &= \delta(\delta^{\gamma} V_{\mu}^{r}) \\
&= \delta(\delta^{\gamma}) V_{\mu}^{r} + \delta^{\gamma} \delta(V_{\mu}^{r})
\end{aligned}$$

and by identification we can see that

$$\delta V_{\mu}^{r} = \xi^{\hat{\rho}} \partial_{\hat{\rho}} V_{\mu}^{r} + \partial_{\mu} \xi^{\nu} V_{\nu}^{r}$$

wich is exactly the same transformation as a vector with Lie derivative [4] Now using the same method for the other block we get for the $\hat{V}_{\alpha}^{a}$

$$\delta \Phi_{\alpha}^{a} = \xi^{\rho} \partial_{\rho} \Phi_{\alpha}^{a}$$

wich is exactlty the transformation of a scalar field. Finnaly for the V_{μ}^{a} we got

$$A_{\mu}^{\alpha} = \xi^{\rho} \partial_{\rho} A_{\mu}^{\alpha} + \partial_{\mu} \xi^{\rho} A_{\rho}^{\alpha} + \frac{\partial_{\mu} \xi^{\alpha}}{2\kappa}$$

wich is also the transformtion of a vector field plus a correction terme at the end. moreover another one might see is that if we variate ϵ^{α} we got

$$\delta V_{\mu}^{r} = \delta \Phi_{\alpha}^{a} = 0, \tag{14a}$$

$$\delta A_{\mu}^{\alpha} = \frac{\partial_{\mu} \xi^{\alpha}}{2\kappa}. \tag{14b}$$

so what rest of the D-dimensional ξ^{α} restricted to y-independent transformation is a set of E abelian Gauge invariance

Now we need to calculate somme spin connexion to discover which one let's rewrite the action by block of non-hatted index that gave us

$$\begin{aligned}
S = -\frac{1}{4\kappa^2} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \Big[& \omega^{r}{}_{st} \omega^{st}{}_{r} + \omega^{r}{}_{sc} \omega^{sc}{}_{r} + \omega^{r}{}_{bt} \omega^{bt}{}_{r} + \omega^{r}{}_{bc} \omega^{bc}{}_{r} + \omega^{a}{}_{st} \omega^{st}{}_{a} + \omega^{a}{}_{sc} \omega^{sc}{}_{a} + \omega^{a}{}_{bt} \omega^{bt}{}_{a} + \omega^{a}{}_{bc} \omega^{bc}{}_{a} \\
& + \omega^{r}{}_{rt} \omega^{t}{}_{s}{}^{s} + \omega^{r}{}_{rt} \omega^{t}{}_{b}{}^{b} + \omega^{r}{}_{rc} \omega^{c}{}_{s}{}^{s} + \omega^{r}{}_{rc} \omega^{c}{}_{b}{}^{b} + \omega^{a}{}_{at} \omega^{t}{}_{s}{}^{s} + \omega^{a}{}_{at} \omega^{t}{}_{b}{}^{b} + \omega^{a}{}_{ac} \omega^{c}{}_{s}{}^{s} + \omega^{a}{}_{ac} \omega^{c}{}_{b}{}^{b} \Big]
\end{aligned} \tag{15}$$

now to minimize the number of connexion we have to calculate we will use the fact that

$$\omega_{rst} = -\omega_{rts} \tag{16}$$

furthermore on can with any kind of updown index do the next manipulation

$$\omega_{rs}{}^{t} = \eta^{tn} \omega_{rsn} = -\eta^{tn} \omega_{rns} = -\omega_{r}{}^{t}{}_{s}$$

therefore we can rewrite the action with only 6 spins componants being.

$$S = -\frac{1}{4\kappa^2}\int d^Dx \int \frac{d^Ey}{\rho(E)}\hat{V}\Big[\omega^r{}_{st}\omega^{st}{}_r + \omega^r{}_{sc}\omega^{sc}{}_r + \omega^r{}_{bt}\omega^{bt}{}_r + \omega^r{}_{bc}\omega^{bc}{}_r + \omega^a{}_{st}\omega^{st}{}_a + \omega^a{}_{sc}\omega^{sc}{}_a + \omega^a{}_{bt}\omega^{bt}{}_a + \omega^a{}_{bc}\omega^{bc}{}_a \\ + \omega^r{}_{rt}\omega^t{}_s{}^s + \omega^r{}_{rt}\omega^t{}_b{}^b + \omega^r{}_{rc}\omega^c{}_s{}^s + \omega^r{}_{rc}\omega^c{}_b{}^b + \omega^a{}_{at}\omega^t{}_s{}^s + \omega^a{}_{at}\omega^t{}_b{}^b + \omega^a{}_{ac}\omega^c{}_s{}^s + \omega^a{}_{ac}\omega^c{}_b{}^b\Big] \tag{17}$$

4.4 finalisation

5 reduction of the new fields

5.1 scalar field reduction

5.2 Two form fiel reduction

6 Observation of new symmetry

7 Conclusion

References

[1] Joel Scherk and John H Schwarz. “How to get masses from extra dimensions”. In: *Nuclear Physics B* 153 (1979), pp. 61–88.

[2] Albert Einstein. “Zur allgemeinen relativitätstheorie”. In: *Sitzungsber. preuss. Akad. Wiss* 44.2 (1915), pp. 778–786.

[3] Th Kaluza. “Zum unitätsproblem der physik”. In: *Sitzungsber. Preuss. Akad. Wiss. Berlin (Math. Phys.)* 1921.arXiv: 1803.08616 (1921), pp. 966–972.

[4] Daniel Z Freedman and Antoine Van Proeyen. *Supergravity.* Cambridge university press, 2012.

[5] Jnanadeva Maharana and John H Schwarz. “Noncompact symmetries in string theory”. In: *Nuclear Physics B* 390.1 (1993), pp. 3–32.

A lagrangian

By using the principle of least action and the variational principle we can see that

$$\begin{aligned}\delta S &= \delta\int \mathcal{L}dx^\mu \\ &= \delta\int \frac{\partial\mathcal{L}}{\partial\eta_\rho}\delta\eta_\rho + \frac{\partial\mathcal{L}}{\partial\partial_\nu\eta_\rho}\delta\partial_\nu\eta_\rho dx^\mu \\ &= \delta\int \frac{\partial\mathcal{L}}{\partial\eta_\rho}\delta\eta_\rho - \partial_\nu\frac{\partial\mathcal{L}}{\partial\partial_\nu\eta_\rho}\delta\eta_\rho + \underbrace{\partial_\nu\Big(\frac{\partial\mathcal{L}}{\partial\partial_\nu\eta_\rho}\delta\eta_\rho\Big)}_{\text{border terme equal to zero}} dx^\mu \\ &= 0\end{aligned}$$

which gave the equation

$$\boxed{\frac{\partial\mathcal{L}}{\partial\eta_\rho} - \partial_\nu\frac{\partial\mathcal{L}}{\partial\partial_\nu\eta_\rho} = 0} \tag{18}$$

which is in fact the equation of mouvement for the field